Nama: Abdul Rozak A
NIM: 221344033
Kelas: 2B-TNK
Rangkuman 9 Elektronika Digital Lanjut

1. Random Counter Sinkron

Random Counter Sinkron merupakan counter sinkron tetapi urutan angkanya tidak menentu. Contohnya seperti Mod 4 dengan urutan 2 3 1 0 menggunakan JK-FF maka:

STATE TABLE

DESIMAL	OUTPUT				INPUT			
	PRESENT STATE		NEXT STATE					
	Q_1	Q_0	Q_1	Q_0	J_1	K_1	J_0	K_0
2	1	0	1	1	d	0	1	d
3	1	1	0	1	d	1	d	0
1	0	1	0	0	0	d	d	1
0	0	0	1	0	1	d	0	d

K-Map J_1

$Q_1 \backslash Q_0$	0	1
0	1	0
1	d	d

$J = \overline{Q_0}$

K-Map K_1

$Q_1 \backslash Q_0$	0	1
0	d	d
1	0	1

$K_1 = Q_0$

K-Map J_0

$Q_1 \backslash Q_0$	0	1
0	0	d
1	1	d

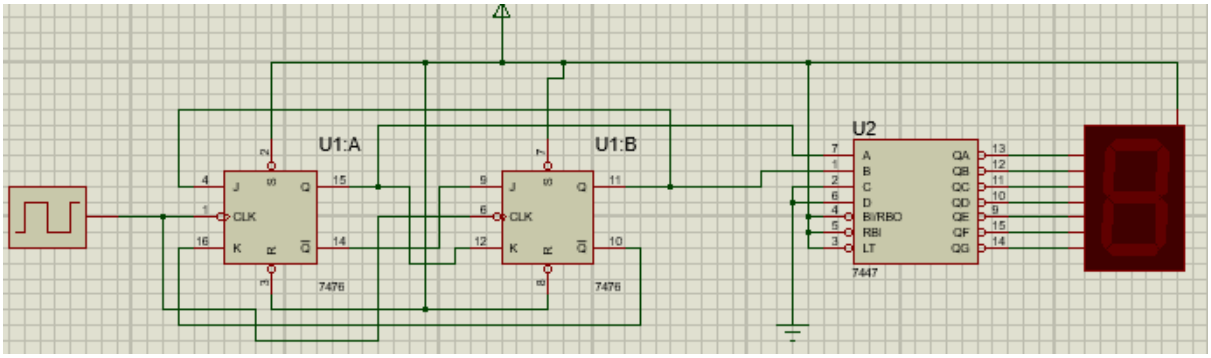
$J_0 = Q_1$

K-Map K_0

$Q_1 \backslash Q_0$	0	1
0	d	1
1	d	0

$K_0 = \overline{Q_1}$

RANGKAIAN



Rangkaian *Random Counter* Sinkron

1. Rangkaian *Random Counter* dari Modulus 4

Contoh Rangkaian Random Counter : 0, 6, 8, 15

Modulus 4 Sinkron :

Tabel 2. 1 Tabel Modulus 4 Sinkron

Q_1	Q_0
0	0
0	1
1	0
1	1

Random Counter yang diinginkan :

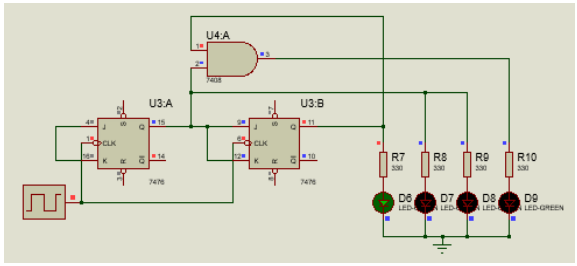
Tabel 2. 2 Tabel Random Counter 0, 6,8,15

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	1	1	0
1	0	0	0
1	1	1	1

Sehingga dapat diketahui :

- Q0 = Q1.Q3
- J1 dan J2 = 1
- J3 = Q2

Maka Rangkaiannya :



Skema Rangkaian Random Counter : 0, 6, 8, 15